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Improved infrared spectroscopic discrimination between gall bladder (GB) polyps and GB cancer using component-descriptive spectral features of separated phases from bile

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This study demonstrates a unique strategy for enhancing infrared (IR) spectroscopic discrimination between gall bladder (GB) polyps and cancer. This strategy includes the separation of raw bile juice into three sections of organic, aqueous, and amphiphilic phases and a cooperative combination of all IR spectral features of each separated phase for the discrimination. Raw bile juice is viscous and complex in composition because it contains fatty acids, cholesterol, proteins, phospholipids, bilirubin, and other components; therefore, the acquisition of IR spectra providing more component-discernible information is fundamental for improving discrimination. For this purpose, raw bile juice was separated into an aqueous phase, mostly containing bile salts, an organic phase with isolated lipids, and an amphiphilic phase, mainly containing proteins. The subsequent IR spectra of each separated phase were mutually characteristic and complementary to each other. When all the IR spectral features were combined, the discrimination was improved compared to that using the spectra of raw bile juice with no separation. The cooperative integration of more component-specific spectra obtained from each separated phase enhanced the discrimination. In addition, the IR spectra of the major constituents in bile juice, such as bile acids, conjugated bile salts, lecithin, and cholesterol, were recorded to explain the IR features of each separated phase.

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Introduction

Infrared (IR) and Raman spectroscopy have been extensively studied as simple and fast analytical tools for the identification of human diseases including various cancers using diverse specimens such as blood and tissue.[1–14] The wide adoption of vibrational spectroscopy in the field of biomedical diagnosis is attributed to its ability to provide rich spectral information on molecular structures, the composition of constituents in a sample, and fast data acquisition with minimal or no sample preparation. Since specimens obtained from human subjects are mostly complex in composition, their corresponding component peaks can severely overlap each other and become broad, losing component specificity. Therefore, chemometrics-based multivariate analysis has been predominantly employed to extract relevant spectral signatures buried in feature-indistinct spectra for discriminant analysis aiming to differentiate normal and disease-contracted states.[6–15]

As part of the research on the vibrational spectroscopy-based diagnosis of disease, our group reported the feasibility of IR spectroscopy for discrimination between gallbladder (GB) polyps and stones.[16] Bile juice was a versatile specimen in that study because its composition was potentially representative of the pathological conditions in the GB of patients; however, the composition of bile juice is considerably complex. It is known that 85% of bile juice is water, and the major constituents of the remainder are bile salts (61%), fatty acids (12%), cholesterol (9%), proteins (7%), phospholipids (3%), and bilirubin (3%).[17] When samples with a complex composition such as bile juice are subjected to analysis, the use of IR spectra containing more component-specific information is the most straightforward approach for enhancing sample discrimination, eventually leading to better differentiation of various GB diseases. Nonetheless, this approach was not readily explored in a previous study, in which the transmission IR spectra of dried raw bile juice samples containing all the components were directly fed into the discriminant analysis.

To improve the IR spectroscopic discrimination of GB-related diseases, a strategy of separating bile juice into organic,

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aqueous, and amphiphilic phases and combining all IR features of each of the separated phase for final analysis is proposed in this study. By this separation, components extracted from each of the phases are expected to differ according to polarity, and the subsequent composition will become obviously simpler compared to that of raw bile juice. Therefore, IR spectra collected from each separated phase could be more component-characteristic and disease-descriptive. Moreover, IR information of the separated phases would be mutually complementary to each other, so the combination of all IR features is potentially synergetic to enhance discrimination.

To evaluate the feasibility of the proposed strategy, IR spectroscopic discrimination between GB polyps and GB cancer has been attempted in this study. Even though GB cancer does not occur at high frequency (the incidence of about 1.5% among total cancers), it has a dismal prognosis, with a 5-year survival rate of 7.4%.[18,19] Long-term survival can be expected only if appropriate surgical resection is accomplished in an early stage, so the early detection of GB cancer, especially against GB polyps, is important from a clinical viewpoint. Currently, several imaging-based methods[20–24] such as sonography, computerized tomography (CT), and magnetic resonance imaging (MRI) are available for this purpose. Of these methods, CT and MRI provide better diagnostic performance due to the use of contrast agents. Nonetheless, the distinction between GB polyps and GB cancer by physical appearance in the obtained images is not sufficiently accurate, since the decision is largely dependent on the size of polypoid lesions.[23–25] Although a biopsy of epithermal cell tissue could provide a final diagnosis, a supportive and non-invasive method that is able to improve the accuracy of diagnosis by incorporating chemical information would be valuable. In this context, IR spectroscopy is a potential candidate and, thereby, employed here for further exploration.

Initially, the IR spectra of raw bile juice obtained from relevant patients were acquired by drop-and-dry measurements, as performed in a previous study. Next, each sample of raw bile juice was mixed with a $CHCl_3 : CH_3OH$ (2 : 1) solution and separated into three sections consisting of organic, aqueous, and amphiphilic phases. Then, the IR spectra of the separated phases were acquired by the same procedure, and the resulting spectral features were compared with those of raw bile juice samples. To characterize the IR bands of each separated phase, the IR spectra of the major constituents in bile juice, such as bile acids (cholic acid, chenodeoxycholic acid, deoxycholic acid, and lithocholic acid), conjugated bile salts (glycodeoxycholate, glycochenodeoxycholate, glycocholate, taurodeoxycholate, taurochenodeoxycholate, and taurocholate), lecithin, and cholesterol, were analyzed. Next, principal component analysis (PCA) was performed using the IR spectra of raw bile juice and each separated phase, and the subsequent distributions of principal component (PC) scores corresponding to GB polyps and GB cancer were examined to qualitatively assess the two-group discriminability. Then, the potential improvement in the discrimination when combining all IR spectra obtained from the separated phases was investigated.

Experimental section

Collection of bile juice and separation into three different phases

Bile juice was obtained from patients with GB polyps and GB cancer after cholecystectomy at Hanyang University Hospital (Seoul, Korea) and immediately stored in a freezer (below −40 °C) to prevent further denaturation. A total of 16 bile juice samples were obtained from 10 GB polyp and 6 GB cancer patients. The pictures of 6 selected bile juice samples are presented in Fig. 1(a). The sample colors varied from brown to deep green, and no systematic color differences between GB polyps and GB cancer were observed. Written informed consent was obtained from all the patients, and the overall study was performed according to protocols approved by the Institutional Review Board of Hanyang University (HYI-16-229-3). All pure components used in this study were purchased from well-known suppliers (cholesterol, glycodeoxycholate, glycochenodeoxycholate, glycocholate, taurodeoxycholate, taurochenodeoxycholate, taurocholate: Sigma-Aldrich; cholic acid, chenodeoxycholic acid, deoxycholic acid, lithocholic acid: Tokyo Chemical Industry; lecithin: Daejung Chemicals & Metals, Korea).

For the extraction and separation of raw bile juice, each sample (30 μL) was mixed with a $CHCl_3 : CH_3OH$ (2 : 1) solution (600 μL). After vigorous mixing and centrifugation for more than 15 minutes at 9800 RPM, the amphiphilic phase mostly containing proteins settled at the bottom (Fig. 1(b)). After the removal of the amphiphilic phase, the remaining sample was centrifuged again for more than 5 minutes at 2000 RPM. This resulted in the separation of the organic (bottom) and aqueous (top) phases, as shown in Fig. 1(c). The bottom organic phase containing fatty acid, cholesterol, and phospholipid was carefully separated to obtain the aqueous phase containing mostly dissolved bile salts.

Acquisition of IR spectra and spectral preprocessing

For drop-and-dry measurements of the samples, an octadecyltrichlorosilane (OTS)-coated Si wafer with a strong hydrophobic surface was used as a platform on which uniform circular droplets with high contact angles were formed. Before the acquisition of the IR spectra, each organic and aqueous phase

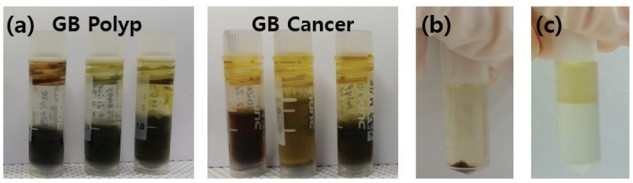

Fig. 1 The pictures of 6 selected raw bile juice samples obtained from 3 gallbladder (GB) polyp and 3 GB cancer patients (a), raw bile juice mixed with $CHCl_3 : CH_3OH$ (2 : 1) solution after vigorous mixing and centrifugation (b), and the same sample containing only the aqueous (top) and organic (bottom) phases (c) after the removal of the settled amphiphilic proteins.

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was diluted 1.5 times with $CHCl_3$ and 2 times with water, respectively. The amphiphilic phase was re-dispersed in water through sonication. Each sample (3 μL) was then dropped on the OTS-coated Si wafer and dried in a desiccator under vacuum conditions for 30 minutes.

The IR spectra of the dried samples were acquired using a Bruker LUMOS FT-IR spectrometer equipped with a microscope and mercury cadmium telluride (MCT) detector. Transmission measurement was accomplished for each sample, and a subsequent spectrum in the 1800–600 cm^{-1} range was acquired by scanning 32 times with a resolution of 4 cm^{-1}. The aperture size of the microscope was 100 μm in diameter. The baseline of the spectrum was linearly corrected at 1800 and 600 cm^{-1}, and each baseline-corrected spectrum was normalized to a maximum absorbance of 1.0 for the peak with the highest intensity. Baseline correction, normalization, and PCA were conducted using MATLAB version 2016a (The Math Works Inc., MA, USA).

The IR spectra of lecithin, cholesterol, glycodeoxycholate, glycochenodeoxycholate, glycocholate, taurodeoxycholate, taurochenodeoxycholate, and taurocholate were similarly collected by a drop-and-dry method. For the measurements, lecithin and cholesterol were dissolved in $CHCl_3$, and 6 bile salt samples were dissolved separately in water : CH_3OH solution. Then, each solution was dropped on the OTS-coated Si wafer and dried for spectral acquisition. The IR spectra of 4 bile acids (cholic acid, chenodeoxycholic acid, deoxycholic acid, and lithocholic acid) in a powder form were directly recorded in attenuated total reflection (ATR) mode using the same IR instrumentation.

Results and discussion

IR spectral features of raw bile juice and separated organic phase samples

Initially, the IR spectral features of raw bile juice samples acquired from patients with GB polyps and GB cancer were examined without further extraction. Fig. 2(a) shows the average spectra of all raw bile samples. The average spectrum of GB cancer was intentionally offset for clear comparison. The shaded areas in each average spectrum correspond to the standard deviations of absorbance at each wavenumber, indicating the degree of spectral variation among the samples. As seen, the overall spectral features of these two average spectra are generally similar to each other; however, the band shapes and intensities at several locations are somewhat dissimilar. At a glance, the intensities of the 1737 cm^{-1} band (corresponding to C=O stretching in the esters of lipids and triglycerides[26,27]) in the two spectra differ, with greater absorption with GB polyps. In addition, the relative band intensities between the 1647 and 1604 cm^{-1} bands are considerably dissimilar in these cases. As reported, these two bands are related with protein structures, for example, an amide I band at 1647 cm^{-1}.[27–29] In addition, the absorption of the peaks in the 1320–1000 cm^{-1} range, including the 1170 cm^{-1} peak originating from the CO–O–C anti-symmetric stretching of the phospholipid,[30,31] is greater for the GB polyp.

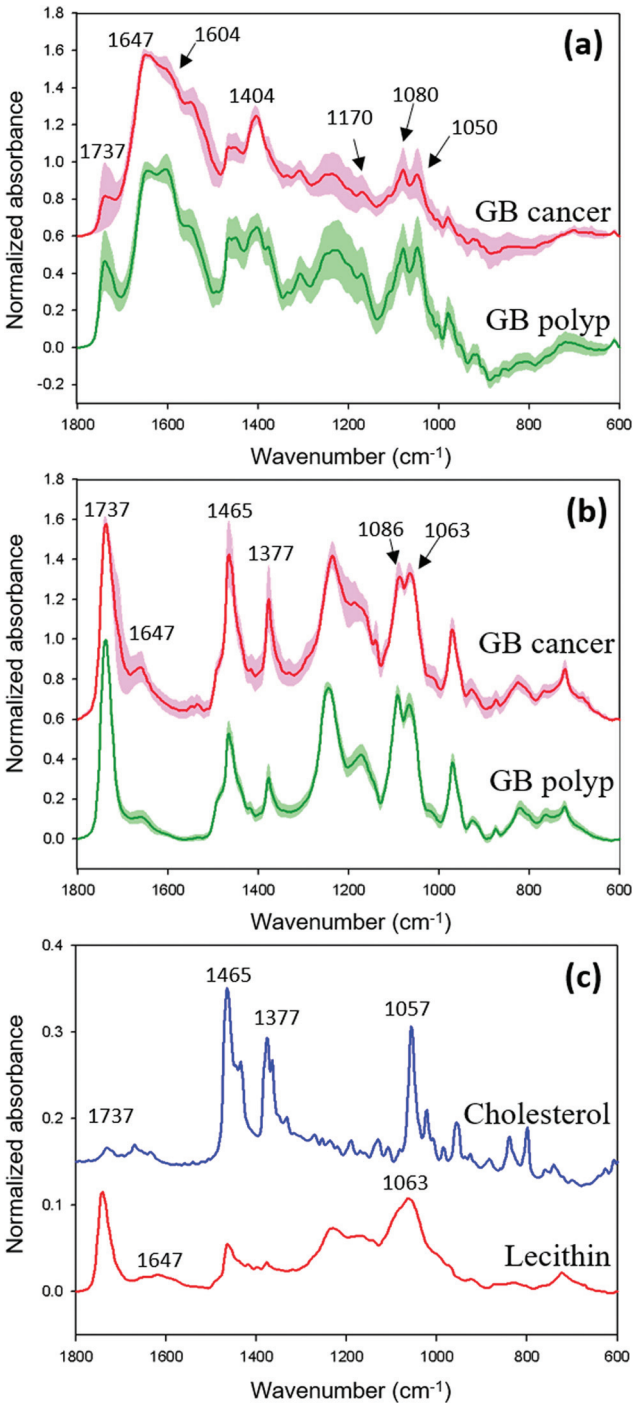

Fig. 2 Average spectra of gallbladder (GB) polyps and GB cancer obtained from the spectra of raw bile juice (a) and organic phase (b) samples. The shaded superimposed area corresponds to the standard deviation of absorbance at each wavenumber. The IR spectra of lecithin and cholesterol are also shown (c).

Since the IR spectral features are substantially overlapping due to the complex composition of the raw bile juice, it is difficult to pinpoint the origin of the spectral difference.

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Nonetheless, based on the observation of the average spectra, it is reasonable to postulate different compositions and/or structures of lipids and proteins in the polyp and cancer samples. However, the sample-to-sample spectral variations are quite large, as indicated by the shaded area; therefore, a clear distinction between the GB polyps and GB cancer will not be realized.

The extraction of components in raw bile juice according to polarity could result in simpler sample compositions in separated phases and more component-characteristic IR spectral features. Fig. 2(b) shows the average IR spectra of the organic phase GB polyp and GB cancer samples with standard deviations (shaded area). The overall shapes of the spectra are distinctly different from those of raw bile juice samples. For example, the 1737 cm^{-1} bands (C═O stretching in the esters of lipids and triglycerides) are dominant due to the extraction of lipids into the organic phase. To support the observed spectral features, the IR spectra of phosphatidylcholine (lecithin) and cholesterol, two of the major hydrophobic components in bile juice, were compared (refer to Fig. 2(c)). The molecular structures of lecithin and cholesterol are shown in Fig. 3. The 1465 and 1377 cm^{-1} bands assignable to –C–H scissoring and –C–H symmetric bending,[31,32] respectively, are dominant in the cholesterol spectrum whereas the 1737 cm^{-1} band is very strong in the lecithin spectrum. It is interesting to note that the combined spectral shape of lecithin and cholesterol is similar to that of the average spectra, indicating the presence of hydrophobic components including lecithin and cholesterol in the organic phase.

When the two average spectra are compared (Fig. 2(b)), there are some differences in band shapes. The width of the 1737 cm^{-1} band in the case of GB cancer is broader than that of the GB polyp, as is the variation at the right side of the band. The magnitudes of the 1465 and 1377 cm^{-1} bands are larger in the case of GB cancer. In addition, the band shapes of the GB polyps and GB cancer in the 1300–1130 cm^{-1} range are fairly dissimilar, and the relative intensities between the 1086 and 1063 cm^{-1} bands differ. The overall observation indicates that the compositions and/or structures of the extracted hydrophobic components in the two sample groups are moderately different, and the variation is greater for GB cancer. Again, differentiation of the two groups would be feasible using the IR spectra of the organic phase samples; however, a clear distinction is difficult to achieve.

IR spectral features of aqueous and amphiphilic phase samples

Fig. 4(a) shows the average IR spectra of the aqueous phase GB polyp and GB cancer samples with designated standard deviations (shaded area). The features of the overall spectra more closely resemble those of raw bile juice samples in comparison with those of the organic phase samples. The resemblance originates from the dominant water-soluble bile salts in the aqueous phase as the most abundant constituents in raw bile

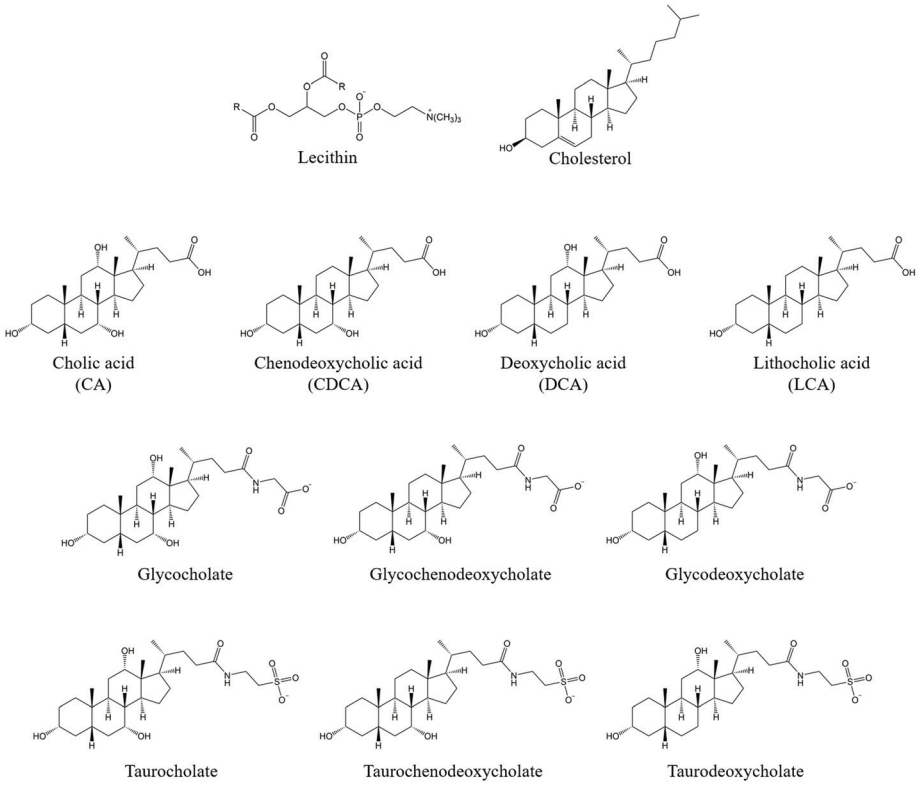

Fig. 3 The molecular structures of selected lipid components, bile acids, and bile salts in this study.

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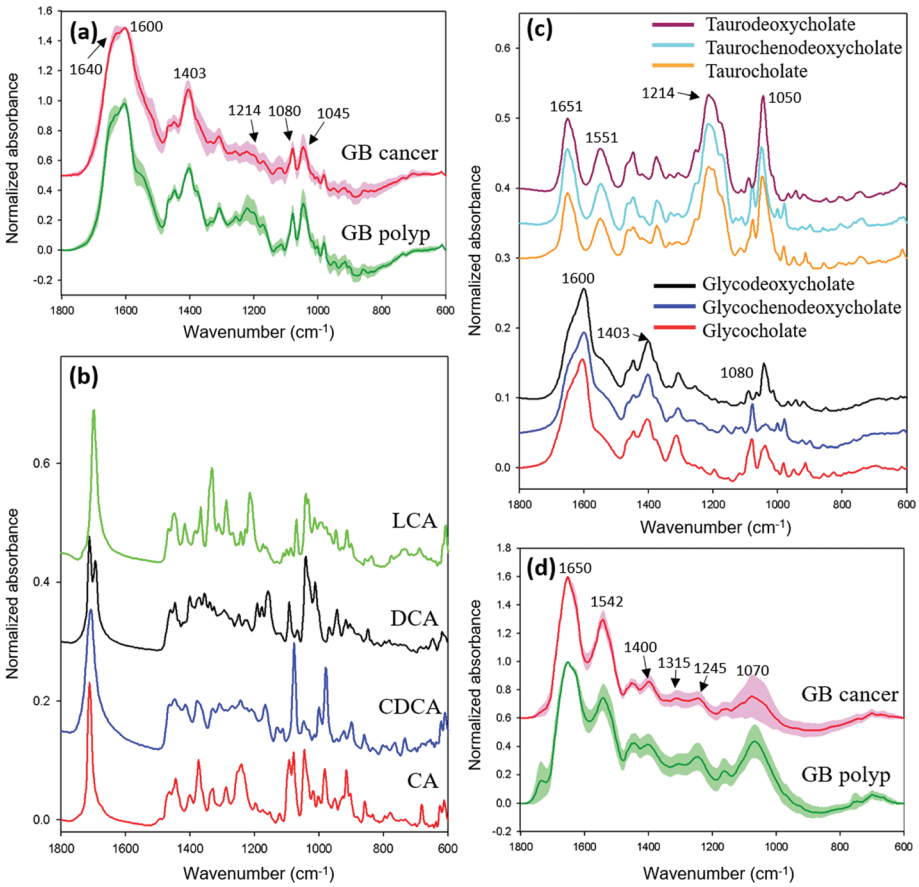

Fig. 4 Average spectra of gallbladder (GB) polyp and GB cancer samples calculated from the spectra of the aqueous (a) and amphiphilic phases (d). The shaded superimposed area corresponds to the standard deviation of absorbance at each wavenumber. The IR spectra of 4 selected bile acids (b) and 6 selected bile salts (c) are also shown.

juice. Therefore, the overall band shapes are largely determined by bile salts. One interesting observation is the absence of the 1737 cm^{-1} band corresponding to C═O stretching in the esters of lipids, since there are no lipids in the hydrophilic aqueous phase.

To explain further the spectral features, the IR spectra of 4 major bile acids were examined as supportive information. Fig. 4(b) shows the IR spectra of cholic acid (CA), chenodeoxycholic acid (CDCA), deoxycholic acid (DCA), and lithocholic acid (LCA), which are derivatives of cholesterol, and their molecular structures are shown in Fig. 3. Primary bile acids (CA and CDCA) are converted into secondary bile acids (CDCA and DCA) by the microbiome in humans.[33–36] The spectral features of bile acids are generally similar to each other due to analogous molecular structures, while their band shapes in the 1500–800 cm^{-1} range show a greater difference. However, the spectra of the bile acids do not resemble those of the aqueous phase samples.

As reported previously, a bile acid is typically conjugated with either glycine or taurine, and the glycine- or taurine-conjugated forms (namely, bile salts) are present in bile juice. The concentration of glycine-conjugated bile salts is three times higher than that of taurine-conjugated bile salts, at approximately 73% and 27%, respectively.[34,36,37] Since bile salts are dominant components in the aqueous phase, the IR spectra of three glycine-conjugated bile salts (glycodeoxycholate, glycochenodeoxycholate, glycocholate) and three taurine-conjugated bile salts (taurodeoxycholate, taurochenodeoxycholate, taurocholate) were compared (refer to Fig. 4(c) and 3 for their molecular structures). The band shapes of the bile salts are different from those of bile acids due to the introduction of glycine or taurine into the bile acid, and the appearances of the IR spectra of glycine-conjugated bile salts are more similar to those of aqueous samples (Fig. 4(a)). This confirms the greater abundance of glycine-conjugated bile salts in the aqueous phase, as described above.

When the average IR spectra of the aqueous phase GB polyp and GB cancer samples are compared, their appearances are similar in general; nonetheless, some notable differences are observed. Initially, the absorbances of peaks around 1214 cm^{-1} are considerably larger in GB polyps. Incidentally, the 1214 cm^{-1} bands are strong in the spectra of taurine-conjugated bile salts but nearly absent in the spectra of glycine-conjugated bile salts. Therefore, it is presumable that the amount of taurine-conjugated bile salts is greater in GB polyps. Next, the shapes of bands in the 1650–1590 cm^{-1}

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range, such as the relative band intensity at 1640 and 1600 cm^{-1}, are different. Overall, there are moderate spectral differences between the aqueous phase of the GB polyp and GB cancer samples, most probably originating from the dissimilar compositions of the bile salts.

Fig. 4(d) shows the average IR spectra of the amphiphilic phase in GB polyp and GB cancer samples. Again, the shaded area corresponds to the standard deviations of absorbance variations. The overall spectral features largely resemble those of protein, based on the observations of protein spectra reported in former studies.[29,38,39] A well-known amide I band at 1650 cm^{-1}, corresponding to C═O stretching in the protein secondary structure, and an amide II band at 1542 cm^{-1}, associated with C–N stretching and N–H bending in protein side chains, are apparent. In addition, several bands at 1400, 1315, 1245, and 1164 cm^{-1} originating from the vibrations of the diverse functional groups in amino acids are also observed.[38,39] Incidentally, the shapes of the spectra are substantially different from those of raw bile juice samples (Fig. 2(a)), since protein is not a major component (roughly 7% among all the components in bile juice).

The average spectra of the amphiphilic phase in the GB polyp and GB cancer samples are fairly dissimilar to each other. The 1737 cm^{-1} band is absent in the spectrum of GB cancer, while its appearance is moderate in the GB polyp spectrum. In addition, the shapes of the amide I and II bands are different in the two cases. The absorption of the 1070 cm^{-1} band corresponding to the PO_2^- vibration of phosphorylated protein[29,31] is considerably larger in the GB polyp, whereas the corresponding intensity variation among the samples is much greater in GB cancer.

On the whole, IR spectra obtained from the separated samples (organic phase, aqueous, and amphiphilic phases) are mutually different due to the separation of components by polarity, and their spectral features are expected to be more component-specific due to the decreased compositional complexity. In addition, the differences in IR spectral features between GB polyps and GB cancer in each case were individually characteristic. Therefore, the combination of more diverse component-descriptive IR spectral information acquired from the separated phases rather than employing sole IR information of a raw bile juice would potentially help to enhance the discrimination between GB polyps and GB cancer.

Evaluation of the discrimination of GB cancer using PCA

To simultaneously investigate the characteristics of all IR spectra of GB polyp and GB cancer samples together and to assess discriminability in each case, PCA was performed separately using the relevant spectra. Fig. 5(a), (b), and (c) show the two-dimensional (2D) scatter plot of 1st *vs.* 2nd scores, 1st *vs.* 3rd scores, and 2nd *vs.* 3rd scores, respectively, constructed using the IR spectra of all raw bile juice samples. The numbers in parentheses are the percentages of spectral variation explained by the corresponding PC over the total spectral variation in the spectral dataset. The squares and filled black circles correspond to the GB polyp and GB cancer samples, respectively, and the numbers inside the symbols designate the corresponding sample numbers. In the 1st *vs.* 2nd score scatter plot, it seems that two groups are present, while their distinction is unclear. Moreover, the #5 polyp sample is located deeper inside the cancer group. The identification of two groups is not realizable in the 1st *vs.* 3rd score and the 2nd *vs.* 3rd score domains. Overlap of the two groups was expected because the sample-to-sample spectral variations in each group were large, as indicated by the shaded areas in Fig. 2(a).

Fig. 5(d)–(f) show the 2D score scatter plots obtained using the IR spectra of the organic phase samples. As is observable from the 1st *vs.* 2nd score and the 1st *vs.* 3rd score scatter plots, the GB polyp and GB cancer groups become more apparent than those employing the IR spectra of raw bile juice. Noticeably, the #5 polyp sample moves closer to the polyp group, while the #1 cancer sample is now located within the polyp cluster. Again, two-group differentiation is difficult to observe in the 2nd *vs.* 3rd score domain. It is clear that the use of IR information of the isolated hydrophobic components in the organic phase is helpful in improving the differentiation of the two groups, but there is still considerable overlap in the middle.

When the IR spectra of the aqueous phase samples were used in the analysis (Fig. 5(g)–(i)), discrimination between the two groups degraded in comparison with that using the IR spectra of the organic phase samples. As shown in the 1st *vs.* 2nd and the 1st *vs.* 3rd score scatter plots, the #1 and #6 cancer samples are within the polyp group, and the #5 and #8 polyp samples are within the territory of the cancer group. Again, the differentiation is not realized in the 2nd *vs.* 3rd score domain. It is reasonable to deduce that the compositions of bile salts in the polyp and cancer samples are moderately different, as found in Fig. 4(a); however, their differences are not substantial enough to clearly indicate a two-group distinction.

The discrimination is most distinct when using the IR spectra of the amphiphilic phase samples containing proteins (refer to Fig. 5(j)–(l)). The large difference in the intensity of the absorption band such as the 1070 cm^{-1} band (PO_2^- vibration of phosphorylated protein) between the two groups (refer to Fig. 4(d)) is responsible for the better two-group identification. Nonetheless, the #6 cancer sample is considerably different from the other cancer samples and too close to the polyp group. Interestingly, in the organic phase samples, the #1 cancer sample is within the polyp group, as discussed.

Based on the observations of all the results above, the use of the separated samples rather than raw bile juice for IR measurement was helpful in improving the discrimination. The spectroscopic information obtained from the amphiphilic phases was relatively more discriminant, while the spectral features obtained from the organic and aqueous phases would be individually complementary as well as potentially synergetic in improving the discrimination. Therefore, to explore the potential, the IR spectra of each separated phase were serially merged and subjected to PCA. The results are presented in Fig. 5(m)–(o). As shown in the 1st *vs.* 2nd and the 1st *vs.* 3rd score scatter plots, discrimination of the two groups becomes

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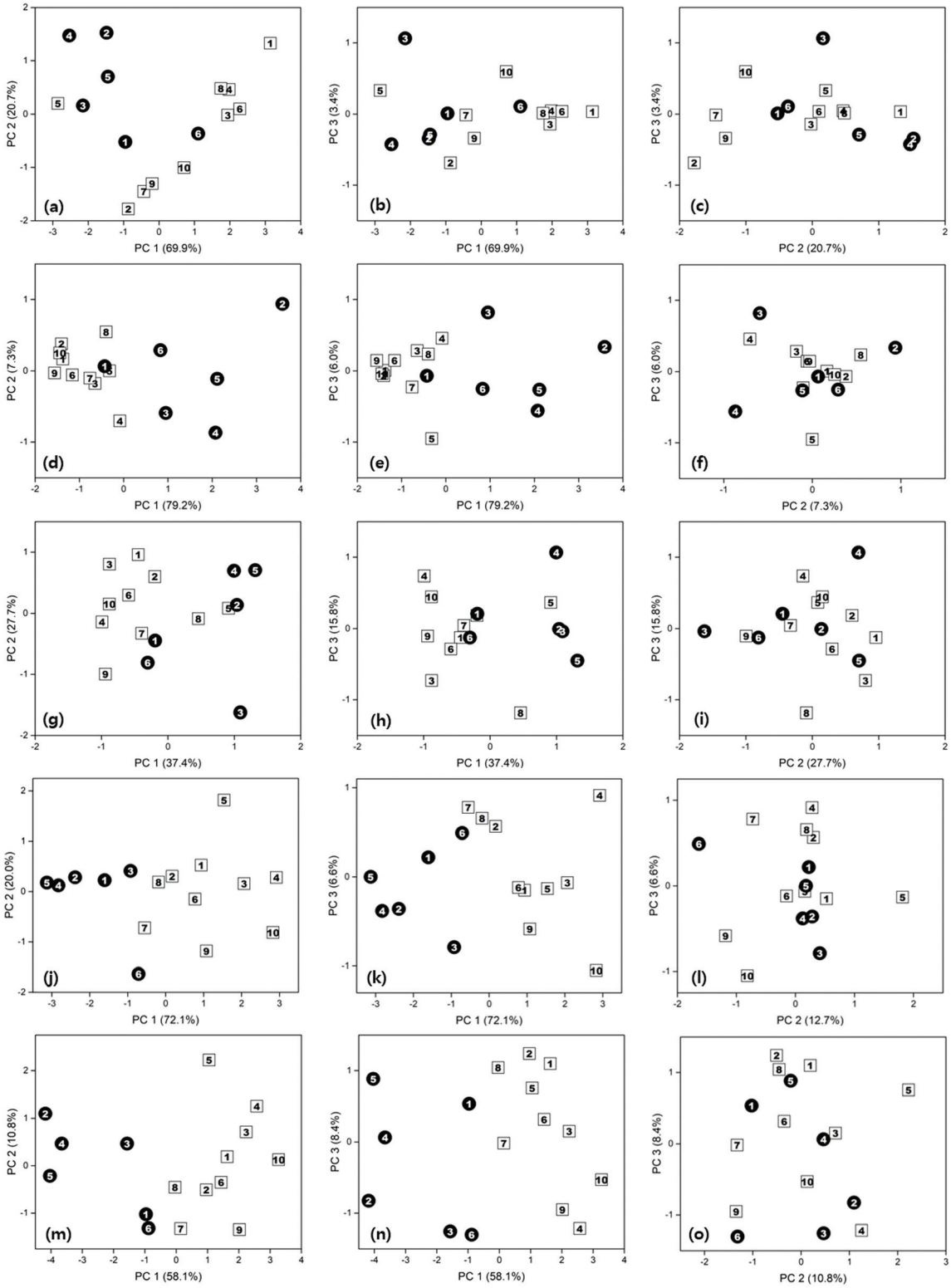

Fig. 5 Two-dimensional scatter plots of the 1st *vs.* 2nd scores (left column), the 1st *vs.* 3rd scores (middle column), and the 2nd *vs.* 3rd scores (left column) obtained from the IR spectra of raw bile juice (1st row), organic phase (2nd row), aqueous phase (3rd row), and amphiphilic phase (4th row) samples. Two-dimensional scatter plots constructed using the merged spectra of the organic, aqueous, and amphiphilic phases are also shown (5th row). The squares and filled black circles correspond to gallbladder (GB) polyp and GB cancer samples, respectively.

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most apparent. The use of only the first scores possibly allows the polyp and cancer groups to be discriminated, although the #1 and #6 cancer samples are relatively close to the polyp group. The simultaneous incorporation of complementary and more component-distinguishable spectral features obtained from each separated phase was the origin for the improved differentiation in the PCA score domain.

Conclusion

The separation of raw bile juice into organic, aqueous, and amphiphilic phases and the combination of IR spectral features acquired from each of the separated phases were effective at enhancing the discrimination between the GB polyp and GB cancer because the spectral features of the separated phases were more component-discernible and cooperatively complementary in the discrimination. Although the number of samples used in this study is not sufficient to justify real clinical applicability, we are confident of the potential of the proposed strategy as a means to enhance the discriminability of major GB-related diseases. We are presently collecting a large number of bile juice samples, which is time-consuming, for future analysis to assess the clinical utility by evaluating quantitative discrimination accuracy with statistical significance. In parallel, effective feature extraction methods that are able to select important target-descriptive variables in combined IR spectra without including redundant and noisy information will be developed. Ultimately, this will be accompanied by an analysis of the full components in bile juice using LC-MS, which is necessary for a fundamental understanding of sample discrimination.

Conflicts of interest

There are no conflicts to declare.

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